Corrigé — Exercice 7

1) $\det(A) = 2(6-4) - 5(2-2) + 3(2-3) = 4 - 0 - 3 = 1 \neq 0$: A inversible.

2) On calcule $A^2 = \begin{pmatrix} 12 & 31 & 22 \\ 7 & 18 & 13 \\ 6 & 15 & 11 \end{pmatrix}$ et $A^3 = \begin{pmatrix} 77 & 197 & 142 \\ 45 & 115 & 83 \\ 38 & 97 & 70 \end{pmatrix}$.

Alors $A^3 - 7A^2 + 4A = \begin{pmatrix} 77-84+8 & 197-217+20 & 142-154+12 \\ 45-49+4 & 115-126+12 & 83-91+8 \\ 38-42+4 & 97-105+8 & 70-77+8 \end{pmatrix} = I_3$.

De $A(A^2 - 7A + 4I) = I$ on tire $A^{-1} = A^2 - 7A + 4I = \begin{pmatrix} 2 & -4 & 1 \\ 0 & 1 & -1 \\ -1 & 1 & 1 \end{pmatrix}$.

3) $X = A^{-1}\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 & -4 & 1 \\ 0 & 1 & -1 \\ -1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ -1 \\ 4 \end{pmatrix}$.

$$(x, y, z) = (-3,\ -1,\ 4)$$